

Mode-I Thesis Master Execution Checklist

Do-not-deviate gate sheet for the Adaptive Remeshing / Phase-Field Fracture thesis

Supervisor-aligned working revision - 3 September 2026

Governing rule: Understand and close the basic Pandey-Kumar Mode-I benchmark before increasing complexity. Every action must map to the active gate and an original proposal task.

Master gate dashboard

Done	Gate	Proposal	Date / initials
☐	Gate 0 - Source/scope freeze	Tasks 1-2	
☐	Gate 1 - Conventional Mode-I reference	Task 3	
☐	Gate 2 - Multi-quantity convergence	Task 3	
☐	Gate 3 - MISESRI mechanism	Task 4 prerequisite	
☐	Gate 4 - Native Python refinement	Task 4	
☐	Gate 5 - Nominal 1% discrepancy	Tasks 4-5	
☐	Gate 6 - Refined reproduction	Task 5	
☐	Gate 7 - ABAQUSER integration	Task 6	
☐	Gate 8 - Higher-complexity benchmarks	Task 7	
☐	Gate 9 - Meshing recommendations	Task 8	
☐	Gate 10 - Future-user documentation	Task 9	

Done	Gate	Proposal	Date / initials
☐	Gate 11 - Final thesis synthesis	Task 10	

Gate 0 - Source and scope freeze

Done	Checklist item	Evidence	Date
☐	Formal 30 June proposal is in the project package and the ten original tasks are unchanged.		
☐	Molnar 2017, Msekh 2015, Pandey-Kumar 2025, and Diddige-Roth-Kiefer 2025 are indexed by role.		
☐	Active scientific question is written in one sentence.		
☐	Mode-II and new topology-changing restart work are explicitly marked HOLD.		
☐	Any post-meeting result is treated as evidence to audit, not a reason to skip gates.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gate 1 - Conventional Mode-I reference (Task 3)

Done	Checklist item	Evidence	Date
☐	Geometry: 1 x 1 mm; sharp 0.5 mm edge crack / seam; Mode-I displacement loading.		
☐	$E = 210 \text{ GPa}$, $\nu = 0.3$, $l_0 = 0.0075 \text{ mm}$, $G_c =$		

Done	Checklist item	Evidence	Date
	2.7e-3 kN/mm verified in input and report.		
☐	Reference mesh topology/seeding documented.		
☐	Complete F-u curve extracted on common units and sign convention.		
☐	Digitized paper target and fixed-mesh reproduction shown together.		
☐	Any uncertainty in digitized reference values is stated.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gate 2 - Multi-quantity convergence qualification

Done	Checklist item	Evidence	Date
☐	Peak force F_{max} evaluated.		
☐	Displacement at peak $u(F_{max})$ evaluated.		
☐	Initial structural stiffness $K_0 = dF/du$ in the initial elastic range evaluated and method documented.		
☐	Full F-u curve error metric computed (not only peak comparison).		
☐	Phase-field/damage contours compared at matched displacements.		

Done	Checklist item	Evidence	Date
☐	Phase-field/profile or crack-path convergence assessed separately from force convergence.		
☐	Energy status addressed: qualified internal UEL energy unavailable unless explicitly implemented; external-work integral, if used, labeled correctly.		
☐	Element count, increments, cutbacks, CPU/walltime and memory recorded.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gate 3 - MISESRI mechanism verification

Done	Checklist item	Evidence	Date
☐	MISESRI defined as recovered-stress discretization error indicator, not phase-field/damage error.		
☐	Figure 3.1 data confirmed as element-field extraction; plotting location/centroid convention documented.		
☐	Correct pre-analysis step and output position verified.		
☐	Correct standard/facsimile element set verified.		

Done	Checklist item	Evidence	Date
☐	Spatial regions of high MISES/ERI identified before viewing the refined mesh.		
☐	Expected refinement pattern written as a hypothesis.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gate 4 - Native Python refinement implementation (Task 4)

Done	Checklist item	Evidence	Date
☐	RemeshingRule call audited line-by-line.		
☐	adaptiveRemesh uses the intended ODB.		
☐	Mesh changes genuinely; before/after topology/count recorded.		
☐	No generateMesh or reseeding call overwrites the adaptive result.		
☐	Refined connectivity converted into the required co-located UEL and companion UMAT layers.		
☐	Nodes, connectivity, sets, BCs, properties, element types, and seam topology pass deterministic checks.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gate 5 - Nominal 1% discrepancy audit

Done	Checklist item	Evidence	Date
☐	Literature value 13,941 finite elements recorded with exact citation.		
☐	Current nominal 1% result 71,320 recorded as a discrepancy, not normalized away.		
☐	2% and 5% cases labeled sensitivities only; no claim that literature used 2%.		
☐	Coarse mesh topology and seed sizes checked.		
☐	Pre-analysis loading/increment history checked.		
☐	Remeshing set/region checked.		
☐	errorTarget, refinementFactor, min/max size and no-coarsening settings checked.		
☐	Abaqus release/version tested only if justified and isolated.		
☐	Outcome classified: verified cause OR unresolved because publication does not specify enough detail.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gate 6 - Refined reproduction (Task 5)

Done	Checklist item	Evidence	Date
☐	Refined solver reaches intended displacement endpoint or failure is scientifically diagnosed.		
☐	Full F-u curve compared with fixed reference and digitized paper curve.		
☐	Fmax, upeak and K0 compared.		
☐	Matched d contours/profiles compared.		
☐	Crack path compared.		
☐	Mesh density and h/10 resolution documented.		
☐	Cost compared only after equivalence assessment.		
☐	Final conclusion distinguishes reproduction quality from preprocessing discrepancy.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gate 7 - ABAQUSER integration (Task 6)

Done	Checklist item	Evidence	Date
☐	Authentic IMFD ABAQUSER executable/script or documented interface obtained.		
☐	Integration procedure		

Done	Checklist item	Evidence	Date
	captured reproducibly.		
☐	Visualized state variables checked against independent extraction.		
☐	If the tool remains unavailable, supervisor decision and exact dependency boundary recorded.		
☐	No internal companion bridge is silently relabeled as the authentic IMFD ABAQUSER tool.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gate 8 - Higher complexity (Task 7)

Done	Checklist item	Evidence	Date
☐	Mode-I gates 1-7 closed or supervisor explicitly authorizes exception.		
☐	Mode-II benchmark physics and expected crack path defined before running.		
☐	Each sensitivity study changes one variable with a predeclared hypothesis.		
☐	Global force, field/path and energy/work evidence evaluated separately.		
☐	State-transfer numerical diffusion is quantified before repeated mapping		

Done	Checklist item	Evidence	Date
	claims are made.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Gates 9-11 - Recommendations, documentation, thesis

Done	Checklist item	Evidence	Date
☐	Task 8 recommendations linked to verified sensitivity evidence and stated applicability range.		
☐	Task 9 package contains .inp, .for, .py, commands.txt, environment notes and provenance.		
☐	Supervisor can rerun the model without receiving multi-GB ODB files.		
☐	Task 10 thesis structure follows scientific logic rather than historical chronology.		
☐	Every major claim is tagged mentally as known, verified numerically, or unresolved.		
☐	Open questions remain visibly open; no mechanism is invented to make the narrative look complete.		

Gate decision: ☐ PASS ☐ HOLD ☐ REWORK Reviewer/initials: _____

Reason / next smallest action: _____

Pre-job anti-deviation card

- ☐ Which original proposal task does this job advance? Task ____
- ☐ Which active gate does it close? Gate ____
- ☐ Scientific question / hypothesis: _____
- ☐ Single intended change from the previous qualified case: _____
- ☐ What must remain frozen? _____
- ☐ Primary outputs required: _____
- ☐ Acceptance criteria declared before submission: _____
- ☐ If the result is unexpected, the next action will be the smallest isolated diagnostic - not a new branch.